



Unknown source in spatially distributed systems: Identifiability analysis and estimation[☆]

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ABSTRACT

Point source estimation aims to detect and localize a concentrated diffusive source as well as to estimate its intensity and induced field from pointwise-in-time-and-space measurements of sensors spread over the area to be monitored. The space–time dynamics of the diffused field is modeled by an *advection–diffusion–reaction partial differential equation* (ADR PDE) and a *finite element* (FE) method is adopted in order to spatially discretize the ADR PDE model. *Source identifiability*, i.e. the ability to detect the source and uniquely identify its location and intensity, is analyzed in a system-theoretic framework by providing sufficient conditions in terms of rank tests on suitable polynomial matrices. Further, a novel *finite element multiple model* (FE-MM) filtering approach to source estimation is presented. The approach consists of running a bank of FE Kalman filters, each conditioned to the source being placed in a given element of the FE mesh, and then combining the estimates of such filters in order to produce estimates of the source location and intensity as well as of the diffusing field. The effectiveness of the proposed source estimation algorithm is demonstrated via simulation experiments in both cases of motionless source of unknown position and mobile source.

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1. Introduction

The detection and localization of diffusive sources (e.g. of heat, polluting agents, toxic biochemical substances, etc.) has recently attracted great interest within both the control and signal processing communities for several reasons including: the availability of low-cost wireless sensor devices measuring the induced field (e.g. temperature, concentration) which can be easily deployed in large number over the area to be monitored; the relevance of such a task in industrial and environmental monitoring, homeland security, situational awareness for a wide range of applications (e.g. fire detection, pollution monitoring, detection and localization of terrorist biochemical attacks, etc.).

This paper focuses on source estimation, referred to as the joint real-time task of detecting the presence of a point field-emitting source, localizing it, estimating its intensity, and monitoring the emitted field. In this respect, source estimation algorithms existing in the literature essentially follow two mainstream approaches:

- a first approach models the source-emitted field in steady-state, thus ignoring its transient time evolution, and therefore results into a parametric (static) estimation problem. Within this approach, it is possible to either construct an optimization problem where the source state is estimated via minimization of a cost function (Matthes et al., 2005; Waterschoot & Leus, 2012), or adopt the Bayesian inference method (Bourne et al., 2019; Hazart et al., 2014; Ristic et al., 2015; Xue et al., 2018; Zhao & Nehorai, 2007) to approximately compute the posterior of the source state and then find the source state estimate via *expected a posteriori* (EAP) or *maximum a posteriori* (MAP) criteria. It is worth pointing out that, for slowly diffusing sources, the steady-state approach can imply a very long, possibly unsatisfactory, detection/localization delay;
- conversely, the second approach explicitly takes into account the space–time field dynamics and, thus, amounts to a state (dynamic) estimation problem. Such a problem

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can be solved via either Kalman-like methods (Demetriou, 2007; Ram & Veeravalli, 2007; Rossi et al., 2004; Weimer et al., 2009) or sequential Monte Carlo methods (Hutchinson et al., 2019a, 2019b; Park & Oh, 2020; Ristic et al., 2019), where the latter require large computational load since a large number of particles are normally employed so as to enhance the estimation accuracy. For a comprehensive literature review on source state estimation, the reader is referred to Hutchinson et al. (2017).

In order to provide faster detection/localization of slowly diffusing sources, this paper will follow the latter, i.e., the dynamic approach. In particular, the space-time field dynamics is described by an *advection-diffusion-reaction partial differential equation* (ADR PDE) that accounts for the presence of an unknown point source of unknown intensity and position responsible for emitting the field of interest. The *finite element* (FE) method is then exploited for spatial discretization of the ADR PDE. After time-discretization, the original infinite-dimensional initial-boundary value problem is, therefore, transformed into a finite-dimensional discrete-time linear system, of possibly very high order, with state vector consisting of the field values in the vertices of the FE mesh, input vector representing the source intensity, and input matrix depending on the source location. In this framework, this paper provides two major contributions to the source estimation problem:

- first, inspired by the classic notion of structural identifiability (Bellman & Astrom, 1970; Grewal & Glover, 1976), this work defines the concept of *source identifiability*, i.e. the possibility of detecting the source and uniquely determining its position and intensity from available pointwise-in-time-and-space field measurements. Specifically, system-theoretic conditions for identifiability are derived in terms of rank tests on suitable polynomial matrices for both cases in which the source intensity is regarded as an unknown input or is modeled as the output of an appropriate exosystem;
- second, a multiple-model Kalman filtering approach (Bar-Shalom et al., 2001) to source estimation is pursued by considering all hypotheses (modes) corresponding to the source location in any possible element of the FE mesh plus a further hypothesis accounting for the possible source absence. Specifically, we devise an *ad-hoc* multiple-model algorithm, referred to with the acronym FE-MM standing for *Finite Element-Multiple Model*, so as to cope with the peculiarity of the adopted FE model, and then we exploit such algorithm for estimation of both a motionless and a moving source.

This paper builds upon previous work on centralized and distributed dynamic field estimation via finite-element approximation (Battistelli et al., 2015a, 2015b, 2017). Preliminary work on dynamic field estimation in the presence of an unknown source altering the field can be found in Battistelli et al. (2015b). In the present paper, such preliminary work has been significantly expanded, for instance, by: considering reaction, in addition to advection and diffusion, in the PDE field model; proposing a new multiple-model method to perform source estimation; providing proofs of the theoretical source identifiability results; presenting simulation experiments for the newly proposed algorithm relative to both motionless and mobile sources.

The rest of the paper is organized as follows. Section 2 formulates the source estimation problem of interest. Section 3 derives a FE approximation of the original infinite-dimensional source diffusion model. Section 4 is devoted to the source identifiability analysis. Section 5 presents the multiple-model Kalman

filtering approach to source estimation. Section 6 demonstrates the effectiveness of the proposed approach by means of simulation examples. Finally, Section 7 ends the paper with concluding remarks and perspectives for future work.

2. Source estimation problem statement

In this section we are going to mathematically formulate the source estimation problem of interest in this paper. Let us consider linear second-order partial differential equations of the form

$$\frac{\partial x}{\partial t} + \mathcal{L}(x) = f \quad \text{in } \Omega \quad (1)$$

with initial condition

$$x(\mathbf{p}, 0) = x_0(\mathbf{p}) \quad \text{in } \Omega \quad (2)$$

and boundary condition

$$\mathcal{B}(x) = \gamma \quad \text{on } \partial\Omega \quad (3)$$

where: $x(\mathbf{p}, t)$ is the space-time dependent scalar field of interest (e.g. concentration, temperature), defined over the space-time domain $\Omega \times \mathbb{R}$; the space domain Ω is assumed to be bounded with smooth boundary $\partial\Omega$; $\mathbf{p} \in \Omega$ is the d -dimensional ($d \in \{1, 2, 3\}$) position vector; $t \in \mathbb{R}$ denotes time; $\mathcal{L}(\cdot)$ and $\mathcal{B}(\cdot)$ are the *advection-diffusion-reaction* and, respectively, *Robin* operators defined as follows

$$\mathcal{L}(x) = -\lambda \nabla^2 x + \mathbf{v}^\top \nabla x + g x \quad (4)$$

$$\mathcal{B}(x) = \partial x / \partial \mathbf{n} + \beta x \quad (5)$$

where ∇ and ∇^2 denote as usual *gradient* and *Laplacian* operators. When the linear operator $\mathcal{L}$ takes the form (4), (1) is referred to as the *advection-diffusion-reaction* (ADR) equation. In particular, λ is a constant-in-time and spatially-uniform diffusion coefficient; $\mathbf{v}(\mathbf{p})$ is the advection velocity vector; g is the decay rate of the reaction term; $\beta(\mathbf{p}) \geq 0$ is a, possibly space-dependent, coefficient; $\partial x / \partial \mathbf{n} = \mathbf{n}^\top \nabla x$, $\mathbf{n}$ being the outward pointing unit normal vector of the boundary $\partial\Omega$; $\gamma(\mathbf{p}, t)$ is the forcing term acting on the boundary $\partial\Omega$; $f(\mathbf{p}, t)$ is the point source input modeled as

$$f(\mathbf{p}, t) = \begin{cases} 0, & \text{if no source exists} \\ u(t) \delta(\mathbf{p} - \mathbf{p}^0(t)), & \text{otherwise} \end{cases} \quad (6)$$

with unknown *intensity* $u(t)$ and *position* $\mathbf{p}^0(t) \in \Omega$, where $\delta(\mathbf{p} - \mathbf{p}^0)$ denotes the *Dirac delta* centered at $\mathbf{p}^0$. The aim is to detect the source presence and jointly estimate $u(t)$, $\mathbf{p}^0(t)$, $x(\mathbf{p}, t)$ given pointwise-in-space-and-time measurements

$$y_{k,i} = h_i(x(\mathbf{s}_i, t_k)) + v_{k,i} \quad (7)$$

provided by sensors $i \in \mathcal{S} = \{1, \dots, S\}$, located at positions $\mathbf{s}_i \in \Omega$, at discrete sampling instants t_k , $k \in \mathbb{Z}_+ = \{1, 2, \dots\}$, such that $0 < t_1 < t_2 < \dots$, and affected by measurement noises $v_{k,i}$.

The above stated dynamic estimation problem is clearly infinite-dimensional. In order to obtain a tractable estimation problem, a finite-dimensional approximation of the PDE will be considered in the next section.

3. Finite element approximation

In this section we are going to show how to approximate the infinite-dimensional dynamic estimation problem stated in the previous section into a finite-dimensional one by means of a semi-discrete approximation of (1) with respect to the space variable based on the *finite element* (FE) method. Subsequently, a discrete-time linear state-space model will be obtained via total

discretization of the ADR equation based on a finite difference scheme for the time derivative.

The FE method is based on a *weak* or integral formulation of (1). First, multiplying both sides by a generic space-dependent test function $\varphi(\mathbf{p})$ and integrating over the space domain Ω , yields the expression

$$\int_{\Omega} \frac{\partial x}{\partial t} \varphi d\mathbf{p} - \lambda \int_{\Omega} \nabla^2 x \varphi d\mathbf{p} + \int_{\Omega} \mathbf{v}^T \nabla x \varphi d\mathbf{p} + \int_{\Omega} g x \varphi d\mathbf{p} = \int_{\Omega} f \varphi d\mathbf{p}. \quad (8)$$

Next, by applying Green's identity to the diffusive term in (8), the following weak form of the initial-boundary value problem (1)–(3) can be obtained

$$\int_{\Omega} \frac{\partial x}{\partial t} \varphi d\mathbf{p} + \lambda \int_{\Omega} \nabla^T x \nabla \varphi d\mathbf{p} + \int_{\Omega} \mathbf{v}^T \nabla x \varphi d\mathbf{p} + \int_{\Omega} g x \varphi d\mathbf{p} - \lambda \int_{\partial\Omega} \frac{\partial x}{\partial \mathbf{n}} \varphi d\mathbf{p} = \int_{\Omega} f \varphi d\mathbf{p} \quad (9)$$

which, by enforcing the boundary condition (3), leads to

$$\int_{\Omega} \frac{\partial x}{\partial t} \varphi d\mathbf{p} + \lambda \int_{\Omega} \nabla^T x \nabla \varphi d\mathbf{p} + \int_{\Omega} \mathbf{v}^T \nabla x \varphi d\mathbf{p} + \int_{\Omega} g x \varphi d\mathbf{p} - \lambda \int_{\partial\Omega} (\gamma - \beta x) \varphi d\mathbf{p} = \int_{\Omega} f \varphi d\mathbf{p} \quad (10)$$

where $\nabla^T x$ denotes the transpose of the gradient of x .

The key practical ingredient of the FE approximation is the domain discretization, which is carried out by means of a suitable tessellation of Ω , i.e. by subdividing the space domain into a finite set of non overlapping regions, or *elements* $\mathcal{E}_1, \dots, \mathcal{E}_v$, such that $\Omega = \bigcup_{j=1}^v \mathcal{E}_j$. This collection of elements (triangles or quadrilaterals in 2D, polyhedral in 3D) defines a FE mesh with vertices $\mathbf{p}_j \in \Omega, j = 1, \dots, n$. Then, it is possible to define an appropriate set of basis functions $\phi_j(\mathbf{p}), j = 1, \dots, n$, and write an approximation of the unknown field $x(\mathbf{p}, t)$ as

$$x(\mathbf{p}, t) = \sum_{j=1}^n \phi_j(\mathbf{p}) x_j(t) = \boldsymbol{\phi}^T(\mathbf{p}) \mathbf{x}(t) \quad (11)$$

where: $x_j(t)$ is the unknown expansion coefficient of function $x(\mathbf{p}, t)$ relative to time t and basis function $\phi_j(\mathbf{p})$; $\boldsymbol{\phi}(\mathbf{p}) = \text{col}\{\phi_j(\mathbf{p})\}_{j=1}^n$ and $\mathbf{x}(t) = \text{col}\{x_j(t)\}_{j=1}^n$. Then each basis function ϕ_j is a piece-wise polynomial vanishing outside the FEs around $\mathbf{p}_j$ and such that $\phi_j(\mathbf{p}_i) = \delta_{ij}$, where δ_{ij} is the Kronecker delta.

By taking the test function φ equal to the selected basis functions, the Galerkin method (Brenner & Scott, 1996; Pelosi et al., 2009), which approximates the weak form of a PDE in a finite-dimensional subspace, can be applied to (10) so as to obtain

$$\begin{aligned} & \underbrace{\left[\int_{\Omega} \boldsymbol{\phi}(\mathbf{p}) \boldsymbol{\phi}^T(\mathbf{p}) d\mathbf{p} \right]}_{\mathbf{M}} \dot{\mathbf{x}}(t) + \underbrace{\left[\lambda \int_{\Omega} \nabla^T \boldsymbol{\phi}(\mathbf{p}) \nabla \boldsymbol{\phi}^T(\mathbf{p}) d\mathbf{p} \right]}_{\mathbf{S}_{\alpha}} \mathbf{x}(t) \\ & + \underbrace{\left[\lambda \int_{\partial\Omega} \beta(\mathbf{p}) \boldsymbol{\phi}(\mathbf{p}) \boldsymbol{\phi}^T(\mathbf{p}) d\mathbf{p} \right]}_{\mathbf{S}_{\beta}} \mathbf{x}(t) \\ & + \underbrace{\left[\int_{\Omega} \boldsymbol{\phi}(\mathbf{p}) \mathbf{v}^T(\mathbf{p}) \nabla \boldsymbol{\phi}^T(\mathbf{p}) d\mathbf{p} \right]}_{\mathbf{S}_{\nu}} \mathbf{x}(t) \\ & + \underbrace{\left[\int_{\Omega} g \boldsymbol{\phi}(\mathbf{p}) \boldsymbol{\phi}^T(\mathbf{p}) d\mathbf{p} \right]}_{\mathbf{S}_{\mathbf{g}}} \mathbf{x}(t) \end{aligned}$$

$$\begin{aligned} & = \underbrace{\left[\int_{\Omega} \boldsymbol{\phi}(\mathbf{p}) \delta(\mathbf{p} - \mathbf{p}^0) d\mathbf{p} \right]}_{\boldsymbol{\phi}(\mathbf{p}^0)} u(t) \\ & + \underbrace{\left[\lambda \int_{\partial\Omega} \boldsymbol{\phi}(\mathbf{p}) \boldsymbol{\phi}^T(\mathbf{p}) d\mathbf{p} \right]}_{\mathbf{D}} \gamma(t). \end{aligned} \quad (12)$$

Note that for each integral defined on the boundary $\partial\Omega$, it is supposed that the considered functions are the restrictions to $\partial\Omega$ of the original functions defined over Ω , and that an expansion similar to (11) holds also for $\gamma(\mathbf{p}, t)$, such that

$$\gamma(\mathbf{p}, t) = \sum_{j=1}^n \phi_j(\mathbf{p}) \gamma_j(t) = \boldsymbol{\phi}^T(\mathbf{p}) \boldsymbol{\gamma}(t)$$

where $\boldsymbol{\gamma}(t) = \text{col}\{\gamma_j(t)\}_{j=1}^n$. Please notice also that all integrals on the left-hand side of (12) only depend on basis functions and can, therefore, be computed off-line. In particular, these integrals yield the well known *mass* and *stiffness* matrices $\mathbf{M}$ and, respectively, $\mathbf{S} = \mathbf{S}_{\alpha} + \mathbf{S}_{\nu} + \mathbf{S}_{\beta} + \mathbf{S}_{\mathbf{g}}$ that can be easily computed in simple coordinates for first order triangular elements (Pelosi et al., 2009; Silvester & Ferrari, 1996). Thus, the integral equation (12) provides a linear system of ordinary differential equations in the descriptor form

$$\mathbf{M} \dot{\mathbf{x}} + \mathbf{S} \mathbf{x} = \boldsymbol{\phi}(\mathbf{p}^0) u + \mathbf{D} \boldsymbol{\gamma}. \quad (13)$$

Then, we proceed with the discretization in time of (13) with sampling interval δt (i.e. $t_k = k \delta t$), and by approximating the time derivative with the finite difference $\dot{\mathbf{x}}(t) \cong (\mathbf{x}_{k+1} - \mathbf{x}_k)/\delta t$, in order to obtain the fully-discrete approximation of (1) in the form of a discrete-time linear descriptor system

$$\mathbf{M} \left(\frac{\mathbf{x}_{k+1} - \mathbf{x}_k}{\delta t} \right) + \mathbf{S} \mathbf{x}_{k+1} \cong \boldsymbol{\phi}(\mathbf{p}^0) u_k + \mathbf{D} \boldsymbol{\gamma}_{k+1} \quad (14)$$

from which we can finally derive the discrete-time linear (non descriptor) state-space model

$$\mathbf{x}_{k+1} = \mathbf{A} \mathbf{x}_k + \mathbf{B}(\mathbf{p}^0) u_k + \mathbf{b}_k + \mathbf{w}_k \quad (15)$$

where

$$u_k = u(t_{k+1})$$

$$\mathbf{A} = [\mathbf{I} + \delta t \mathbf{M}^{-1} \mathbf{S}]^{-1}$$

$$\mathbf{B}(\mathbf{p}^0) = [\mathbf{I} + \delta t \mathbf{M}^{-1} \mathbf{S}]^{-1} \mathbf{M}^{-1} \delta t \boldsymbol{\phi}(\mathbf{p}^0) \quad (16)$$

$$\mathbf{b}_k = [\mathbf{I} + \delta t \mathbf{M}^{-1} \mathbf{S}]^{-1} \mathbf{M}^{-1} \delta t \mathbf{D} \boldsymbol{\gamma}_{k+1}$$

and $\mathbf{w}_k$ is a process disturbance taking into account also the space-time discretization errors. For a quantitative characterization of such errors, the reader can refer to Becker and Rannacher (2001). Notice that, in Eq. (15), the intensity u_k and the position $\mathbf{p}^0$ of the point source input are unknown and hence must also be estimated along with the state vector $\mathbf{x}_k$. As for the intensity u_k , different models are possible:

- (a) u_k is regarded as an *unknown input* for which no information on the possible time evolution is available (Gillijns & Moor, 2007; Li, 2013);
- (b) u_k is unknown but a dynamic model for its time evolution is available, i.e., it is supposed that u_k is generated as the output of an auxiliary linear system called *exosystem* (Basile & Marro, 1992)

$$\mathbf{q}_{k+1} = \mathbf{F} \mathbf{q}_k + \boldsymbol{\zeta}_k$$

$$u_k = \mathbf{H} \mathbf{q}_k$$

$$(17)$$

where $\mathbf{q}_k$ is the exosystem state and $\boldsymbol{\zeta}_k$ a disturbance input. Here, without loss of generality, it is assumed that the pair $(\mathbf{F}, \mathbf{H})$ is observable.

For instance, if it is known that the unknown intensity u_k can vary slowly with time, its time evolution can be modeled as a random walk by letting $\mathbf{F} = \mathbf{H} = \mathbf{I}$ and taking ζ_k as a zero-mean white noise. Of course, which model is preferable depends on the situation under consideration and, specifically, on available physical insights (if any) on the source intensity.

4. Source identifiability

In this section, an analysis on the possibility of correctly identifying the unknown source location $\mathbf{p}^0$ and intensity u_k is provided both in cases (a) and (b). As usual, the observability/identifiability analysis is carried out in the ideal noise-free case by supposing that the measurements $\mathbf{y}_k$ are generated by

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A}\mathbf{x}_k + \mathbf{B}(\mathbf{p}^0)u_k \\ \mathbf{y}_k &= \mathbf{C}\mathbf{x}_k \end{aligned} \quad (18)$$

where the dimension of $\mathbf{x}_k$ turns out to be the number of vertices of the FE mesh. Accordingly, in case (b), the intensity dynamics will also be assumed noise-free by letting

$$\begin{aligned} \mathbf{q}_{k+1} &= \mathbf{F}\mathbf{q}_k \\ u_k &= \mathbf{H}\mathbf{q}_k. \end{aligned} \quad (19)$$

Notice also that the known input $\mathbf{b}_k$ is not considered in Eq. (18) since, thanks to the superposition principle for linear systems, its contribution is immaterial to the source identification problem.

Let now $\mathbb{S}$ be the linear space of all real-valued sequences on the nonnegative integers $\mathbb{Z}_+$ and let us denote by $\mathcal{U} \subseteq \mathbb{S}$ the set of all possible time-evolutions of the intensity u_k which are consistent with the available model. Clearly, in case (a) we simply have $\mathcal{U} = \mathbb{S}$, while in case (b) $\mathcal{U}$ is the set of all the possible output behaviors of system (19). Let also denote by $\mathbf{y}(\mathbf{x}_0, \mathbf{p}^0, u)$ the output behavior of system (18) when the initial state is $\mathbf{x}_0$, the source location is $\mathbf{p}^0$ and the source intensity evolves according to the sequence u . The following notion can be introduced.

Definition 1. The point source is said to be *identifiable* if

$$\mathbf{y}(\mathbf{x}_0, \mathbf{p}^0, u) \neq \mathbf{y}(\bar{\mathbf{x}}_0, \bar{\mathbf{p}}^0, \bar{u}) \quad (20)$$

for any pair of source locations $\mathbf{p}^0, \bar{\mathbf{p}}^0 \in \Omega$, any pair of initial states $\mathbf{x}_0, \bar{\mathbf{x}}_0$, and any nonzero pair of intensity sequences $u, \bar{u} \in \mathcal{U}$, with $(\mathbf{p}^0, u) \neq (\bar{\mathbf{p}}^0, \bar{u})$.

In words, identifiability of the point source corresponds to the fact that different sources (in terms of location and intensity) always give rise to different output behaviors or, equivalently, corresponds to the invertibility of the mapping from $(\mathbf{p}^0, u)$ to the output sequence $\mathbf{y}$. Notice that, in the above definition, the trivial case in which both u and $\bar{u}$ are excluded, but it is allowed that either u or $\bar{u}$ be zero so as to account for the possibility of distinguishing between presence or absence of the source input.

Since only the observable part of (18) influences the output behavior $\mathbf{y}$, it is convenient to consider an alternative representation of system (18) obtained by means of the Kalman observability decomposition (Fortuna & Frasca, 2012). This amounts to considering an invertible transformation matrix $\mathbf{T}$ such that

$$\begin{aligned} \mathbf{T}^{-1}\mathbf{A}\mathbf{T} &= \begin{bmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \quad \mathbf{T}^{-1}\mathbf{B}(\mathbf{p}^0) = \begin{bmatrix} \mathbf{B}_1(\mathbf{p}^0) \\ \mathbf{B}_2(\mathbf{p}^0) \end{bmatrix} \\ \mathbf{C}\mathbf{T} &= [\mathbf{C}_1 \quad \mathbf{0}] \end{aligned} \quad (21)$$

with observable pair $(\mathbf{A}_{11}, \mathbf{C}_1)$. Of course, when $\mathbf{C}$ is chosen so that $(\mathbf{A}, \mathbf{C})$ is observable, we have $\mathbf{A}_{11} = \mathbf{A}$ and $\mathbf{C}_1 = \mathbf{C}$. All the results of this section will refer to the system (18) or (18)–(19) in the case (b), under the observability decomposition (21). Detailed proofs of all results are provided in the Appendix. By exploiting standard results on observability and left invertibility of linear systems, the following theorem can now be stated.

Theorem 1. For any pair of distinct source locations $\mathbf{p}^0, \bar{\mathbf{p}}^0 \in \Omega$, consider the polynomial matrix

$$\Psi(z, \mathbf{p}^0, \bar{\mathbf{p}}^0) = \begin{bmatrix} z\mathbf{I} - \mathbf{A}_{11} & \mathbf{B}_1(\mathbf{p}^0) & \mathbf{B}_1(\bar{\mathbf{p}}^0) \\ \mathbf{C}_1 & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

with $z \in \mathbb{C}$. Then, the following facts hold:

(i) in case (a), the source is identifiable if and only if

$$\text{rank} \{ \Psi(z, \mathbf{p}^0, \bar{\mathbf{p}}^0) \} = n_o + 2 \quad (22)$$

for any $z \in \mathbb{C}$ and for any $\mathbf{p}^0, \bar{\mathbf{p}}^0 \in \Omega$ with $\mathbf{p}^0 \neq \bar{\mathbf{p}}^0$. Here n_o is the dimension of the observable subsystem of (18);

(ii) in case (b), the source is identifiable if and only if the rank condition (22) holds for any $z \in \text{sp}\{\mathbf{F}\}$ and for any $\mathbf{p}^0, \bar{\mathbf{p}}^0 \in \Omega$ with $\mathbf{p}^0 \neq \bar{\mathbf{p}}^0$, where $\text{sp}\{\mathbf{F}\}$ stands for the spectrum of the matrix $\mathbf{F}$. $\square$

Notice that the only difference between cases (a) and (b) is that in the latter case only the values of z corresponding to eigenvalues of the exosystem have to be considered in the rank test. Notice also that, in this case, the identifiability condition can be rephrased in terms of system gains as follows.

Corollary 1. Consider case (b) and suppose that $\text{sp}\{\mathbf{A}\}$ and $\text{sp}\{\mathbf{F}\}$ are disjoint. Then, the source is identifiable if and only if

$$\text{rank} [\mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}(\mathbf{p}^0), \quad \mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}(\bar{\mathbf{p}}^0)] = 2$$

for any $z \in \text{sp}\{\mathbf{F}\}$ and any $\mathbf{p}^0, \bar{\mathbf{p}}^0 \in \Omega$ with $\mathbf{p}^0 \neq \bar{\mathbf{p}}^0$. $\square$

From Theorem 1 and Corollary 1, it is evident that at least 2 sensors are needed in order to guarantee distinguishability of two source locations $\mathbf{p}^0, \bar{\mathbf{p}}^0$. However, this is just a lower bound since, in general, a larger number of sensors may be needed (see also the analysis below). The main drawback of the derived rank conditions is that they have to be satisfied for any pair of source locations $\mathbf{p}^0, \bar{\mathbf{p}}^0$ belonging to the continuous space domain Ω . This means that, in order to verify whether a given set of sensors ensures source identifiability, an infinite number of conditions have to be checked, which clearly makes the test practically infeasible. A first, approximated, approach to sidestep such a difficulty would amount to restricting the attention only to a finite number of possible source locations, for example corresponding to the vertices $\mathbf{p}_i$ of the FE mesh. A second, more theoretically sound, approach consists in looking for alternative conditions by exploiting the structure of the system matrices resulting from application of the FE method.

With this respect, consider the most typical situation in which the elements are chosen as d -dimensional simplexes (i.e., intervals when $d = 1$, triangles when $d = 2$, or tetrahedrons when $d = 3$) and the basis functions $\phi_j(\mathbf{p})$ are piecewise linear. Further, let $\mathcal{E}_1, \dots, \mathcal{E}_v$ denote the elements of the considered mesh and, for a generic element $\mathcal{E}_j$, let $\mathcal{V}_j \subset \{1, \dots, n\}$ be the set of indices corresponding to the vertices of $\mathcal{E}_j$. Notice that, for a d -dimensional simplex, $\mathcal{V}_j$ contains exactly $d + 1$ elements. Then, in this case, the input matrix $\mathbf{B}(\mathbf{p}^0)$ can be written as a convex combination of the input matrices associated to the vertices of the element $\mathcal{E}_j$ containing $\mathbf{p}^0$, i.e.,

$$\mathbf{B}(\mathbf{p}^0) = \sum_{i \in \mathcal{V}_j} \omega_i(\mathbf{p}^0) \mathbf{B}(\mathbf{p}_i) \quad (23)$$

where $\omega_i(\mathbf{p}^0)$, $i \in \mathcal{V}_j$, are the convex combination coefficients defining the location of the source within the element $\mathcal{E}_j$, i.e., $\mathbf{p}^0 = \sum_{i \in \mathcal{V}_j} \omega_i(\mathbf{p}^0) \mathbf{p}_i$ with $\omega_i(\mathbf{p}^0) \geq 0$ and $\sum_{i \in \mathcal{V}_j} \omega_i(\mathbf{p}^0) = 1$. This makes it possible to derive sufficient conditions for source identifiability which have to be checked only for each pair of distinct elements. To this end, for a generic element $\mathcal{E}_j$, let $\mathbf{B}(\mathcal{E}_j)$ denote the matrix

obtained by row juxtaposition of the column matrices $\mathbf{B}(\mathbf{p}_i)$, for $i \in \mathcal{V}_j$, and let $\mathbf{B}_1(\mathcal{E}_j)$ be obtained in an analogous way from the matrices $\mathbf{B}_1(\mathbf{p}_i)$, $i \in \mathcal{V}_j$. Then, the following result holds.

Theorem 2. Let the input matrices be as in (23) and, for any pair of distinct elements $\mathcal{E}_j, \mathcal{E}_\ell$ of the mesh, consider the polynomial matrix

$$\Psi_{j\ell}(z) = \begin{bmatrix} z\mathbf{I} - \mathbf{A}_{11} & \mathbf{B}_1(\mathcal{E}_j) & \mathbf{B}_1(\mathcal{E}_\ell) \\ \mathbf{C}_1 & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

with $z \in \mathbb{C}$. Then, the following facts hold:

(i) in case (a), the source is identifiable if

$$\text{rank} \{ \Psi_{j\ell}(z) \} = n_o + |\mathcal{V}_j \cup \mathcal{V}_\ell| \quad (24)$$

for any $z \in \mathbb{C}$ and for any $\mathcal{E}_j, \mathcal{E}_\ell$ with $j \neq \ell$;

(ii) in case (b), the source is identifiable if the rank condition (24) holds for any $z \in \text{sp}\{\mathbf{F}\}$ and for any $\mathcal{E}_j, \mathcal{E}_\ell$ with $j \neq \ell$. $\square$

Notice that in condition (24), the term $|\mathcal{V}_j \cup \mathcal{V}_\ell|$ represents the number of distinct vertices in $\mathcal{E}_j \cup \mathcal{E}_\ell$. As a consequence, $|\mathcal{V}_j \cup \mathcal{V}_\ell| \leq 2(d+1)$ where the equality holds if and only if $\mathcal{E}_j$ and $\mathcal{E}_\ell$ have no common vertices. Hence, Theorem 2 suggests that, in the d -dimensional case, $2(d+1)$ sensors may be needed in order to have source identifiability.

Remark 1. The rank condition of Theorem 2 provides a computationally feasible way to verify whether a given set of sensors guarantees source identifiability, since it has to be checked only for a finite number of cases, i.e., for each pair of distinct elements. Additional insights can be gained by recalling that system (18) can also be written as a linear descriptor system (see (14))

$$\begin{aligned} \mathbf{E} \mathbf{x}_{k+1} &= \mathbf{M} \mathbf{x}_k + \delta t \phi(\mathbf{p}^0) u_k \\ \mathbf{y}_k &= \mathbf{C} \mathbf{x}_k \end{aligned} \quad (25)$$

where the matrices $\mathbf{E}$, $\mathbf{M}$ and $\phi(\mathbf{p}^0)$ have very specific structures. In particular, the non-zero elements of the matrices $\mathbf{E}$, $\mathbf{M}$ correspond to connected vertices in the graph associated to the FE mesh. Further, for any mesh vertex $\mathbf{p}_i$, $\phi(\mathbf{p}_i)$ coincides with $\mathbf{e}_i$, the i th vector of the canonical basis. If it is assumed that system (25) is observable, it can be easily shown that in condition (24) the matrix $\Psi_{j\ell}(z)$ can be replaced by

$$\tilde{\Psi}_{j\ell}(z) = \begin{bmatrix} z\mathbf{E} - \mathbf{M} & \phi(\mathcal{E}_j) & \phi(\mathcal{E}_\ell) \\ \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

where $\phi(\mathcal{E}_j)$ denotes the matrix obtained by row juxtaposition of the column matrices $\phi(\mathbf{p}_i) = \mathbf{e}_i$ with $i \in \mathcal{V}_j$. Hence, if it is further assumed that the sensor locations coincide with vertices of the FE mesh, it is possible to relate the rank of the matrix $\tilde{\Psi}_{j\ell}(z)$ to the topology of the FE mesh as well as to the sensor locations. In fact, results on the rank of matrices of the form of $\tilde{\Psi}_{j\ell}(z)$ for systems like (25) defined over graphs have been recently provided in the literature (Pasqualetti et al., 2013, 2015). While such results are only generic (i.e., they hold for almost all the dynamical systems compatible with the graph topology, but counterexamples can exist), nevertheless they provide useful guidelines on where to place the sensors inside the domain Ω . The interested reader is referred to Pasqualetti et al. (2013, 2015) for further details on this issue.

5. Source estimation algorithm

In this section we are going to introduce the algorithm proposed for source estimation and also provide an analysis on its computational complexity. Based on the fact that distinct source locations correspond to different process behaviors, and thanks

to the finite element approximation, the key idea of the proposed source estimation algorithm relies upon the assumption that system (15), at each time step, obeys one of a finite set of ADR models. To this end, the *multiple model* (MM) approach (Bar-Shalom et al., 2001) provides a suitable tool, as it accounts for the uncertainty about the system input location, assuming that the real evolution of the system follows one of the possible modes of operation. In particular, the idea is to match each hypothesis of source being located in a generic element of the mesh, to a distinct operating mode of the system. This makes it possible to run in parallel a finite number of mode-matched Kalman filters, one for each element of the generated mesh. Hence, each mode j associated to the hypothesis that a point source is located in $\mathbf{p}^0$, contained in element $\mathcal{E}_j$, is characterized by the following mode-matched model

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A} \mathbf{x}_k + \mathbf{B}(\mathcal{E}_j) \omega(\mathbf{p}^0) u_k + \mathbf{w}_k, \quad j = 1, 2, \dots, \nu \\ \mathbf{y}_k &= \mathbf{C} \mathbf{x}_k + \mathbf{v}_k \end{aligned} \quad (26)$$

where $\omega(\mathbf{p}^0)$ is the $(d+1)$ -dimensional column vector of coefficients $\omega_i(\mathbf{p}^0)$, $i \in \mathcal{V}_j$, introduced in Section 4. Further, in order to model the case in which no point source is present, an extra *source-free* operating mode needs to be added to the set of possible modes of the MM algorithm. Summing up, including the no-source mode from now on denoted as $j = 0$, the set of possible modes has cardinality $\nu + 1$.

Bearing in mind the previous points about source detection and localization, the additional joint source intensity and state estimation can be carried out by constructing an augmented system for the MM estimator, as the aggregate of the original system (26) and a suitable model for the unknown input time evolution. To this end, let us introduce $\boldsymbol{\eta}_k = \text{col}(\omega_i(\mathbf{p}^0) u_k, i \in \mathcal{V}_j)$ so that $\mathbf{B}(\mathbf{p}^0) u_k = \mathbf{B}(\mathcal{E}_j) \boldsymbol{\eta}_k$. Then, the augmented system for a generic mode j originated from (26), takes the following form for $j = 1, 2, \dots, \nu$

$$\begin{aligned} \begin{bmatrix} \mathbf{x}_{k+1} \\ \boldsymbol{\eta}_{k+1} \end{bmatrix} &= \begin{bmatrix} \mathbf{A} & \mathbf{B}(\mathcal{E}_j) \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x}_k \\ \boldsymbol{\eta}_k \end{bmatrix} + \begin{bmatrix} \mathbf{w}_k \\ \boldsymbol{\zeta}_k \end{bmatrix} \\ \mathbf{y}_k &= [\mathbf{C} \quad \mathbf{0}] \begin{bmatrix} \mathbf{x}_k \\ \boldsymbol{\eta}_k \end{bmatrix} + \mathbf{v}_k \end{aligned} \quad (27)$$

whereas, for $j = 0$

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A} \mathbf{x}_k + \mathbf{w}_k \\ \mathbf{y}_k &= \mathbf{C} \mathbf{x}_k + \mathbf{v}_k. \end{aligned} \quad (28)$$

Note that in (27) the dynamics of the source intensity u_k are assumed to follow a discrete-time random walk. As a result, the joint field and source estimation problem can be reduced to the joint estimation of $\mathbf{x}_k$ and $\boldsymbol{\eta}_k$ for each time step k . Assuming that the source may move within the domain, i.e. the correct operating mode may switch over time, it is convenient to adopt a dynamic MM technique (Bar-Shalom et al., 2001). Whenever the source of interest is assumed motionless, i.e. fixed in an unknown position of the monitored area, the dynamic MM algorithm is, anyway, still applicable by tuning the mode transition probabilities to allow mode jumps only between $j \in \{1, 2, \dots, \nu\}$ and $j = 0$.

Notice that the linear model in (27) does not explicitly take into account the physical constraint that the components of the source vector $\boldsymbol{\eta}_k$ must be non-negative. Hence, in order to prevent negative values in $\boldsymbol{\eta}_k$, special care has to be taken in the MM method to make it suitable for the specific FE modeling adopted in this paper. Alternatively, one could consider estimation approaches that allow for explicitly taking into account constraints in the computation of the state estimates such as moving-horizon estimation (Alessandri & Battistelli, 2020). However, these alternative methods are not pursued here so as to keep the computational load manageable.

Next, an MM algorithm named FE-MM, specifically developed for source estimation by means of the *Finite Element-Multiple Model* problem formulation (27)–(28), is presented. The algorithm refers to a centralized setting wherein the measurements of all sensors are collected and jointly processed in the correction step of each mode-matched filter. For further details on dynamic multiple-model filtering algorithms, the reader is referred to Bar-Shalom et al. (2001).

5.1. FE-MM algorithm for source estimation

The FE-MM algorithm runs a bank of $v + 1$ finite-element Kalman filters matched to the modes j in (27), for $1 \leq j \leq v$, or (28) for $j = 0$. Each filter updates the state estimate, covariance and mode probability relative to mode j by processing the gathered measurements $y_{k,i}$, $i \in \mathcal{S}$. The transitions are modeled by means of a homogeneous Markov chain with known constant transition probabilities

$$\pi_{ij} = \text{prob}(v_k = i | v_{k-1} = j), \quad i, j \in \{0, 1, \dots, v\} \quad (29)$$

where v_k represents the *modal state* (i.e. the mode in operation) at time k . Suitable choices for the transition probabilities will be discussed in Section 6.

For the sake of convenience, we introduce the augmented state vector $\mathbf{x}_k \triangleq [\mathbf{x}_k^\top \boldsymbol{\eta}_k^\top]^\top$. Given the augmented state estimate $\hat{\mathbf{x}}_{k-1|k-1}^j$, the relative covariance $\mathbf{P}_{k-1|k-1}^j$, and the mode probabilities μ_{k-1}^j at time step $k - 1$, then the estimation recursion for mode $j \in \{0, 1, \dots, v\}$ proceeds through the following steps.

1. **Mode-matched filtering:** a mode-matched Kalman filter for each $j \in \{0, 1, \dots, v\}$ carries out the prediction and correction steps, processing the gathered measurements $y_{k,i}$, $i \in \mathcal{S}$. The bank of filters produces mode-conditioned state estimates $\hat{\mathbf{x}}_{k|k}^j$ and covariances $\mathbf{P}_{k|k}^j$ for each mode $j = 1, \dots, v$ via the 2-step (prediction–correction) algorithm detailed below.

(a) Prediction

$$\mathbf{x}_{k|k-1}^j = \begin{bmatrix} \mathbf{A} & \mathbf{B}(\mathcal{E}_j) \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \mathbf{x}_{k-1|k-1}^j \quad (30)$$

$$\mathbf{P}_{k|k-1}^j = \begin{bmatrix} \mathbf{A} & \mathbf{B}(\mathcal{E}_j) \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \mathbf{P}_{k-1|k-1}^j \begin{bmatrix} \mathbf{A}^\top & \mathbf{0} \\ \mathbf{B}(\mathcal{E}_j)^\top & \mathbf{I} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\Sigma}_w & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_\xi \end{bmatrix} \quad (31)$$

where $\boldsymbol{\Sigma}_w$ and $\boldsymbol{\Sigma}_\xi$ denote the covariances of $\mathbf{w}_k$ and ξ_k , respectively.

(b) Correction

$$\mathbf{S}_k^j = [\mathbf{C} \ \mathbf{0}] \mathbf{P}_{k|k-1}^j [\mathbf{C} \ \mathbf{0}]^\top + \mathbf{R}_k \quad (32)$$

$$\mathbf{K}_k^j = \mathbf{P}_{k|k-1}^j [\mathbf{C} \ \mathbf{0}]^\top (\mathbf{S}_k^j)^{-1} \quad (33)$$

$$\mathbf{x}_{k|k}^j = \mathbf{x}_{k|k-1}^j + \mathbf{K}_k^j (\mathbf{y}_k - \mathbf{C} \mathbf{x}_{k|k-1}^j) \quad (34)$$

$$\mathbf{P}_{k|k}^j = (\mathbf{I} - \mathbf{K}_k^j [\mathbf{C} \ \mathbf{0}]) \mathbf{P}_{k|k-1}^j \quad (35)$$

where $\mathbf{R}_k$ is the covariance of $\mathbf{v}_k$.

Please notice that prediction and correction steps (30)–(35) are related to modes $1, \dots, v$ in (27). For mode $j = 0$ in (28), they must be changed accordingly with trivial modifications that are omitted here.

2. **Mode probability update:** Assuming Gaussian measurement noise, mode likelihoods are evaluated as follows

$$\Lambda_k^j = \begin{cases} \mathcal{N}(\zeta_k^j; \mathbf{0}, \mathbf{S}_k^j), & \text{if } \hat{\boldsymbol{\eta}}_{k|k}^j \text{ has no negative entry} \\ \varsigma \cdot \mathcal{N}(\zeta_k^j; \mathbf{0}, \mathbf{S}_k^j), & \text{otherwise} \end{cases} \quad (36)$$

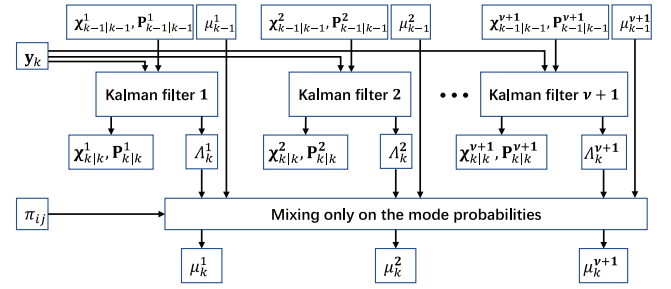


Fig. 1. Flow-chart of the proposed FE-MM algorithm.

where: $\mathcal{N}(\mathbf{x}; \hat{\mathbf{x}}, \boldsymbol{\Sigma}) \propto \exp(-\frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \hat{\mathbf{x}}))$; $\zeta_k^j = \mathbf{y}_k - \mathbf{C} \hat{\mathbf{x}}_{k|k-1}^j$ is the innovation at time k of mode j ; $\mathbf{S}_k^j$ is the associated covariance computed via (32); $\varsigma \in (0, 1)$ is a suitably chosen, typically very small, *likelihood reduction factor* that penalizes source estimates with negative components and, hence, not consistent with the model. Then, mode probabilities are updated by means of the mode likelihoods as follows

$$\mu_k^j = \frac{1}{c_1} \Lambda_k^j \sum_{i=0}^v \pi_{ji} \mu_{k-1}^i \quad (37)$$

where $c_1 = \sum_{j=0}^v \Lambda_k^j \sum_{i=0}^v \pi_{ji} \mu_{k-1}^i$ is the normalization constant. Finally, the source estimate components are projected into the positive octant by setting to zero possible negative entries of $\hat{\boldsymbol{\eta}}_{k|k}^j$.

3. **Source estimation:** Let j^* be the mode with highest probability, i.e.

$$j^* = \arg \max_j \mu_k^j. \quad (38)$$

If $j^* = 0$, no source is detected. Otherwise, the associated mode-conditioned estimate $\hat{\mathbf{x}}_{k|k}^{j^*} \triangleq [(\hat{\mathbf{x}}_{k|k}^{j^*})^\top (\hat{\boldsymbol{\eta}}_{k|k}^{j^*})^\top]^\top$ will be directly used for field monitoring and source estimation. In particular, the field-state estimate can be taken as

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k}^{j^*}. \quad (39)$$

Next, exploiting the structure of the FE approximation, the source location $\mathbf{p}^0$ and intensity u_k can be estimated from $\hat{\boldsymbol{\eta}}_{k|k}^{j^*}$ as follows

$$\begin{aligned} \hat{u}_k &= \sum_{i \in \mathcal{V}_{j^*}} \hat{\eta}_{k|k,i}^{j^*}, \quad \hat{\omega}_i = \frac{\hat{\eta}_{k|k,i}^{j^*}}{\hat{u}_k}, \quad i \in \mathcal{V}_{j^*} \\ \hat{\mathbf{p}}^0 &= \sum_{i \in \mathcal{V}_{j^*}} \hat{\omega}_i \mathbf{p}_i \end{aligned} \quad (40)$$

where $\hat{\eta}_{k|k,i}^{j^*}$ denotes the entry of $\hat{\boldsymbol{\eta}}_{k|k}^{j^*}$ corresponding to vertex i of the element $\mathcal{E}_{j^*}^*$. ■

Remark 2. A flow-chart of the proposed algorithm is shown in Fig. 1. Please notice that, unlike the interactive MM (IMM) algorithm of Bar-Shalom et al. (2001), no mode mixing is applied to state estimates and covariances in the proposed FE-MM algorithm. The reason is that the states of the various elements (modes), which consist of two parts namely the field state $\mathbf{x}$ and the source state $\boldsymbol{\eta}$ in (27), are not mixable as the source states $\boldsymbol{\eta}^j$ relative to different modes j actually specify the source location with respect to different FE elements.

Remark 3. In order to avoid possible numerical problems arising from (40) for small source intensity $\hat{u}_k$ as well as to smoothen source position estimates, adaptive filtering can be applied to directly estimate $\mathbf{p}^0$ without computing the weights $\hat{\omega}_i$. Specifically, we can more robustly estimate $\mathbf{p}^0$ by applying componentwise recursive least-squares (RLS) to the linear regression model

$$\mathbf{z}_k = \hat{u}_k \mathbf{p}^0 + \varepsilon_k \quad (41)$$

where

$$\hat{u}_k = \sum_{i \in \mathcal{Y}_j^*} \hat{\eta}_{k|i,i}^{j*}, \quad \mathbf{z}_k = \sum_{i \in \mathcal{Y}_j^*} \hat{\eta}_{k|i,i}^{j*} \mathbf{p}_i. \quad (42)$$

Such an estimate can be computed recursively by

$$\varrho_k = \frac{\varrho_{k-1}}{\theta + \varrho_{k-1} \hat{u}_k^2} \quad (43)$$

$$\hat{\mathbf{p}}_k^0 = \hat{\mathbf{p}}_{k-1}^0 + \varrho_k \hat{u}_k (\mathbf{z}_k - \hat{u}_k \hat{\mathbf{p}}_{k-1}^0) \quad (44)$$

initialized from variance $\varrho_0 \gg 1$ and estimate $\hat{\mathbf{p}}_0^0 = \mathbf{0}$, $\theta \in (0, 1)$ being an exponential forgetting factor.

Remark 4. This final remark concerns the estimation of a mobile source with the proposed FE-MM algorithm. In the FE-MM algorithm, the source velocity is not explicitly modeled and estimated, while the source position is estimated by linearly combining the vertices of the element corresponding to the matched mode, as it can be seen in (40). Furthermore, it is assumed that the source state $\boldsymbol{\eta}$ does not undergo sharp changes, as its time evolution is modeled in (27) by a random walk. In the case of a fast moving source, a different model for the evolution of the source state, e.g., nearly-constant velocity (Li & Jilkov, 2003) or Ornstein-Uhlenbeck model (Coraluppi & Carthel, 2012), has to be incorporated in the proposed algorithm.

5.2. Computational complexity

Recall that the complexity of the source estimation problem is characterized by n, ν, S, d where: n is the number of vertices of the FE mesh and hence dimension of the field state $\mathbf{x}$; ν is the number of elements of the FE mesh so that the number of required mode-matched Kalman filters is $\nu + 1$; S is the number of deployed sensors and hence dimension of the measurement vector $\mathbf{y}$; $d \in \{1, 2, 3\}$ is the dimension of the spatial domain so that the dimension of the augmented state $\boldsymbol{\chi}$ turns out to be $n + d + 1$. Clearly $d \ll n$ and also $S \ll n$ in practical cases. Hence, the complexity of a single mode-matched Kalman filter is $O(n^3)$, where $O(\cdot)$ stands for “in the order of”, and the overall complexity $O(\nu n^3)$ if the sparsity of the involved matrices in (14) (typically banded) is not exploited. The additional mode mixing step (37), jointly involving all $\nu + 1$ mode-matched filters requires extra $O(\nu^2)$ computation. Hence, the overall computational complexity of the FE-MM algorithm is $O(\max(\nu n^3, \nu^2))$ which is quadratic in ν and cubic in n . Typical FE mesh generation produces more elements than vertices, i.e. $\nu > n$, but actually their numbers are of the same order of magnitude, i.e. $\nu = O(n)$. Hence it can be concluded that the overall computational complexity is $O(n^4)$ while the memory complexity is $O(n^3)$. Of course n , and also ν , depends on both the size of the spatial domain of interest and the required resolution. Let D denote the diameter of the domain Ω and δ the average diameter of the FE cell, then both n and ν are $O((D/\delta)^d)$. In the next section, it will be shown how problems that involve in the order of hundreds FE cells and vertices are feasible for real-time operation, even without exploitation of the sparse structure of the descriptor FE model (14) which is anyway lost in (15) after inversion of the stiffness matrix $\mathbf{M}$. For larger-scale problems and/or applications requiring faster sampling rates, the

computational load of the FE-MM algorithm might become excessive and appropriate remedies should be undertaken like, e.g.: exploiting the sparse nature of the FE model; using recursive filters suitable for problems with a large number of variables like the Ensemble Kalman filter; resorting to Rao-Blackwellized particle filtering (Cheng & Singh, 2007) instead of multiple-model filtering so as to reduce the number of considered hypotheses. These ideas will be investigated in future work.

6. Numerical examples

In this section, results of simulation experiments concerning estimation of both motionless and mobile sources are presented in order to assess performance of the proposed FE-MM algorithm. All the results have been obtained running Matlab[®] 2020 code on a desktop computer with CPU Ryzen 7 5800X @3.80 [GHz], 32 [GB] RAM, and Windows 10 operating system. Here we present the results of numerical experiments concerning a diffusion process governed by the two-dimensional case of (1) with $\mathcal{L}(\mathbf{x}) = -\lambda (\partial^2 x / \partial \xi^2 + \partial^2 x / \partial \eta^2)$ and mixed boundary conditions (3)

$$\partial x / \partial \mathbf{n} + \beta_1 x = g_1 \quad \text{on } \partial \Omega_1 \quad (45)$$

$$\partial x / \partial \mathbf{n} = 0 \quad \text{on } \partial \Omega_2. \quad (46)$$

This model describes, for instance, transient contaminant transport in water bodies (van Genuchten et al., 2013). Parameters $\beta_1 = \frac{\varpi}{\lambda}$ and $g_1 = \frac{\varpi}{\lambda} x_e$ are such that (45) describes an outward/inward diffusive flux across $\partial \Omega_1$ proportional to the concentration difference $x - x_e$ between internal and external environments (external concentration $x_e = 0$ is assumed known, $\varpi = 1$). The homogeneous Neumann boundary condition (46) assumes there is no flux across $\partial \Omega_2$, i.e. it is considered impermeable to the contaminant. Further, (1) implicitly assumes λ constant, here taken as $\lambda = 0.1$.

A sensor network, which consists of 11 nodes, is randomly deployed inside the spatial 2D domain Ω to sample the concentration field of interest, with sampling interval $T_s = 1$ s and standard deviation of measurement noise $\sigma_v = 0.01$. In order to take into account model uncertainties, two triangular meshes are generated over Ω for the finite-dimensional approximation of the monitored field in the considered domain, as depicted in Figs. 2 and 3. The first mesh (85 vertices and 133 elements, see Fig. 2) is used in the implementation of the FE-MM filter, so that 134 mode-matched Kalman filters, 133 for the possible elements and 1 for the no-source hypothesis, are employed. The second, finer, mesh (302 nodes and 532 elements, see Fig. 3) is instead used as ground truth to simulate the behavior of the concentration field and generate the measurements. The simulation for this second mesh runs at a faster sampling interval of 0.1 s. As *true* initial field condition, we consider $\mathbf{x}_0 = \mathbf{0}$, whereas the estimator starts from $\hat{\mathbf{x}}_{1|0} = \mathbf{0}$ with covariance $\mathbf{P}_{1|0} = 100^2 \mathbf{I}$. Moreover, the source intensity estimate is initialized as $\hat{u}_{1|0} = 5$, with associated initial covariance matrix $\mathbf{P}_{1|0}^u = 100^2 \mathbf{I}$, while the *true* average intensity of the source is 15. The standard deviation of the process noise for the simulator is set as $\sigma_w = 0.5$, whereas for the FE-MM filters $\hat{\sigma}_w = 3$. The standard deviation of the intensity increment ζ_k is chosen as $\sigma_\zeta = 0.1$ for the simulator and $\hat{\sigma}_\zeta = 0.8$ for the filters. Moreover, the likelihood reduction factor in (36) is set to $\varsigma = 10^{-8}$, and the RLS parameters of Remark 3 are set to $\theta = 0.9$ and $\rho_0 = 1000$.

Specifically, *root mean square errors* (RMSEs) relative to source position, intensity and source-induced field, are reported as performance indices. These are obtained via comparison of the estimated quantities with the simulated system (*ground truth*).

Next, we consider two separate scenarios, the first concerning a motionless source with unknown position and intensity, and the second a slowly and continuously moving source. The FE-MM approach is validated via simulations. The simulated scenario is illustrated in Fig. 2, wherein two specific cases are considered:

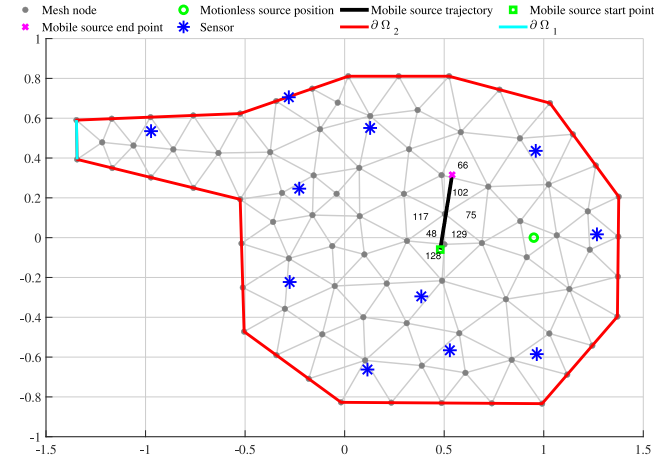


Fig. 2. Simulated scenario, monitored by 11 sensors, for source estimation with mesh (85 nodes, 133 elements) generated over Ω , delimited by boundary $\partial\Omega = \partial\Omega_1 \cup \partial\Omega_2$. Numbers denote element indices.

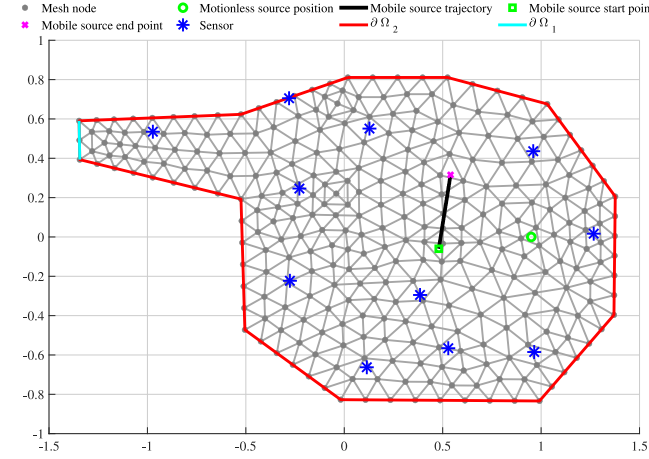


Fig. 3. Simulated ground-truth scenario with finer mesh (302 nodes, 532 elements) generated over Ω , delimited by boundary $\partial\Omega = \partial\Omega_1 \cup \partial\Omega_2$.

- (1) Motionless source, wherein no source is active when the sensing devices start the monitoring activity. After 50 sampling times, a fixed source located at $\mathbf{p}^0 = [0.95, 0]^T$ (marked by the green circle) becomes active, and then is deactivated at time 450, the total simulation time being 500. Since the source is motionless, there is no transition between different elements, and mode jumps can only occur between an element-matched mode and the no-source mode. Accordingly, in this motionless source case-study, jump probabilities in (29) have been taken as $\pi_{ii} = 0.95, \forall i = 0, 1, \dots, v, \pi_{i0} = \pi_{0i} = 0.05, i = 1, \dots, v$, while the remaining ones are set to 0.
- (2) Moving source, wherein the source is activated at time 50 and position $\mathbf{p}^0 = [0.48, -0.06]^T$. After activation, the source keeps motionless in the first 50 sampling intervals, and then it starts to move with velocity 8×10^{-5} and 6×10^{-4} (both toward positive direction) horizontally and vertically, respectively. The motion lasts until time 850, then the source becomes inactive at time 950, the total simulation time being 1000. In contrast to a motionless source, transition probabilities of a mobile source should be designed to reflect the fact that it can move from one element to a neighboring one. In our simulation experiments, jump probabilities (29) for the mobile source have

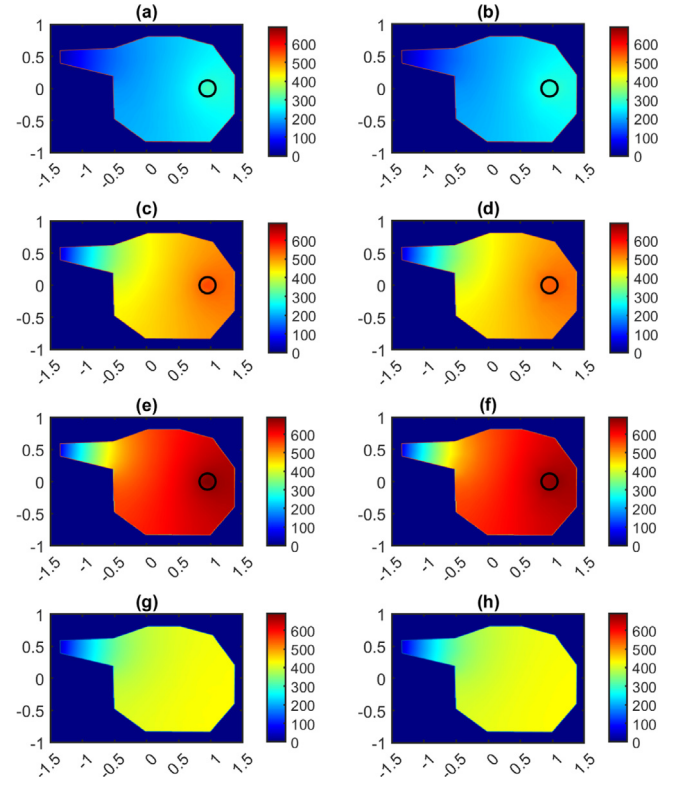


Fig. 4. Field estimation in the presence of a motionless source: true (left) and estimated (right) fields at time steps 100 (a), (b), 200 (c), (d), 350 (e), (f), and 500 (g), (h). The source position is denoted by a black circle.

Table 1
Numerical statistics per iteration.

	A-ET [s]	A-MO [megabyte]
Motionless source	0.1144	8.848
Mobile source	0.1156	8.864

been chosen as $\pi_{ii} = 0.65, \forall i = 0, 1, \dots, v, \pi_{i0} = \pi_{0i} = 0.05, i = 1, \dots, v$, while the remaining 0.3 probability has been equally distributed among all elements $\mathcal{E}_j$ adjacent to $\mathcal{E}_i$.

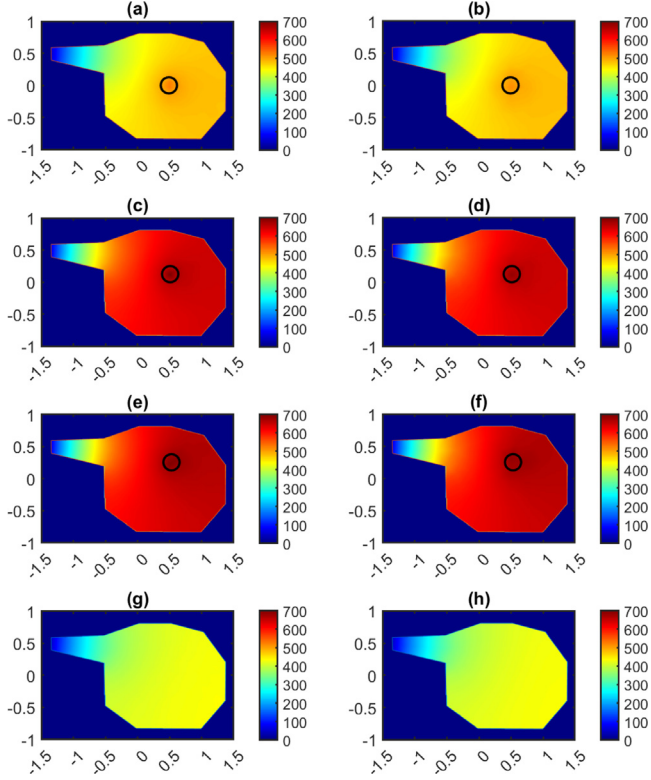
First, we qualitatively show field estimation performance in presence of both motionless and mobile sources in a single Monte Carlo trial. The results are illustrated in Figs. 4 (motionless source) and 5 (mobile source). As it can be seen, the field is correctly estimated in both cases.

Next, we quantitatively illustrate estimation performance by means of RMSEs averaged over 100 Monte Carlo trials. The results are illustrated in Fig. 6, and the probabilities of the selected modes (i.e., the elements wherein the source moves through) in the mobile source estimation case-study are shown in Fig. 7. The average execution time (A-ET) and memory occupation (A-MO) per time recursion are reported in Table 1, where the A-MO is achieved by comparing the difference between *random access memory* (RAM) occupation before and after running the Matlab code. Notice that, even if the code has not been optimized, the execution time is in the order of one tenth of a second thus allowing sampling rates in the order of 10 [Hz], while the average RAM occupation is about 8 [MB] which normally represents a negligible amount of the available memory.

It can be seen that the FE-MM estimator succeeds in both cases to localize the source (Fig. 6(a), (d)) and estimate its intensity (Fig. 6(b),(e)) in a very short time and with remarkable accuracy.

Table 2Performance of the proposed FE-MM method versus process noise variance σ_w .

σ_w	0.5	1	1.5	2	2.5	3	3.5	4
RMSE of source position	0.0086	0.0095	0.0102	0.0107	0.0118	0.0133	0.0149	0.0159
RMSE of source intensity	0.2096	0.2946	0.3429	0.3882	0.4322	0.4896	0.5285	0.5838
RMSE of field state	0.2114	0.2371	0.2692	0.3086	0.3512	0.3991	0.4473	0.4976

**Fig. 5.** Field estimation in the presence of a mobile source: true (left) and estimated (right) fields at time steps 200 (a), (b), 500 (c), (d), 800 (e), (f), and 1000 (g), (h). The source position is denoted by a black circle.

Please notice that the estimation errors of both motionless and mobile sources exhibit visible peaks in correspondence of either source activation or turn-off. This is due to the fact that the probability of the correct mode does not immediately become the largest after source activation/turn-off, so that the extracted mode via MAP criterion (38) is not the correct one. Nevertheless, the FE-MM algorithm promptly recovers once the correct mode has been detected. This demonstrates the effectiveness of the FE-MM source estimator.

Finally, Table 2 shows, for a mobile source, the quantitative performance of the proposed FE-MM method, in terms of RMSE for source position, source intensity and field state estimation, under different values of the process noise standard deviation σ_w for the simulator, setting $\hat{\sigma}_w = 6\sigma_w$ for the FE-MM filters. As it can be seen, source estimation performance degrades for increasing σ_w . Such a result is due to the reduced predictability of the field state which clearly implies larger source position and intensity as well as field errors.

7. Conclusions

The paper has addressed the problem of detecting a diffusive point source and jointly estimating its location, intensity and induced field from pointwise-in-time-and-space field measurements of sensors deployed over the monitored area. This has

been made possible by combining the finite-element method for discretizing in space the diffusion dynamics and the multiple-model Kalman filtering approach. Moreover, source identifiability – namely the possibility of detecting the source existence and uniquely determining its unknown location and intensity – has been analyzed in a system-theoretic framework by providing sufficient conditions in terms of rank tests on suitable polynomial matrices. Possible future work will concern extension of the current one to quickly moving, possibly maneuvering, and/or multiple sources.

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Appendix

Proof of Theorem 1. Let $\mathbf{x}_{1,k} \in \mathbb{R}^{n_0}$ be the state vector of the observable part of (18). Then, we have

$$\begin{aligned}\mathbf{x}_{1,k+1} &= \mathbf{A}_{11}\mathbf{x}_{1,k} + \mathbf{B}_1(\mathbf{p}^0)u_k \\ \mathbf{y}_k &= \mathbf{C}_1\mathbf{x}_{1,k}.\end{aligned}$$

Further, thanks to linearity, the overall output behavior can be decomposed as the sum of the natural (free) response $\mathbf{y}^n(\mathbf{x}_{1,0})$ and the forced response $\mathbf{y}^f(\mathbf{p}^0, u)$. As a consequence, the identifiability condition (20) can be written as $\mathbf{y}^n(\mathbf{x}_{1,0}) + \mathbf{y}^f(\mathbf{p}^0, u) \neq \mathbf{y}^n(\bar{\mathbf{x}}_{1,0}) + \mathbf{y}^f(\bar{\mathbf{p}}^0, \bar{u})$ or equivalently as $\mathbf{y}^n(\mathbf{x}_{1,0} - \bar{\mathbf{x}}_{1,0}) + \mathbf{y}^f(\mathbf{p}^0, u) - \mathbf{y}^f(\bar{\mathbf{p}}^0, \bar{u}) \neq 0$. It is now immediate to see that source identifiability is equivalent to requiring that the system

$$\begin{aligned}\mathbf{x}_{1,k+1} &= \mathbf{A}_{11}\mathbf{x}_{1,k} + \mathbf{B}_1(\mathbf{p}^0)u_k - \mathbf{B}_1(\bar{\mathbf{p}}^0)\bar{u}_k \\ \mathbf{y}_k &= \mathbf{C}_1\mathbf{x}_{1,k}\end{aligned}\quad (47)$$

does not exhibit a zero output trajectory when its input $(u, \bar{u})$ is different from 0. In case (a), when the input trajectories can be arbitrary, this latter condition is clearly equivalent to the left invertibility of system (47). Then, fact (i) follows from well-known results (Basile & Marro, 1992) by noting that $\Psi(z, \mathbf{p}^0, \bar{\mathbf{p}}^0)$ corresponds to the Rosenbrock's system matrix of (47). As for case (b), since both u_k and $\bar{u}_k$ are supposed to be generated by an exosystem of the form (19), we can consider the augmented system resulting from the cascade interconnection of (47) with

$$\begin{aligned}\begin{bmatrix} \mathbf{q}_{k+1} \\ \bar{\mathbf{q}}_{k+1} \end{bmatrix} &= \begin{bmatrix} \mathbf{F} & \mathbf{0} \\ \mathbf{0} & \mathbf{F} \end{bmatrix} \begin{bmatrix} \mathbf{q}_k \\ \bar{\mathbf{q}}_k \end{bmatrix} \\ \begin{bmatrix} u_k \\ \bar{u}_k \end{bmatrix} &= \begin{bmatrix} \mathbf{H} & \mathbf{0} \\ \mathbf{0} & \mathbf{H} \end{bmatrix} \begin{bmatrix} \mathbf{q}_k \\ \bar{\mathbf{q}}_k \end{bmatrix}.\end{aligned}\quad (48)$$

Then, the identifiability condition corresponds to the fact that the autonomous system (47)–(48) does not exhibit a zero output trajectory when its initial condition is different from 0, or in other words to the fact that (47)–(48) is observable. In order to study the observability of (47)–(48), it is convenient to consider an alternative representation of such a system in terms of the

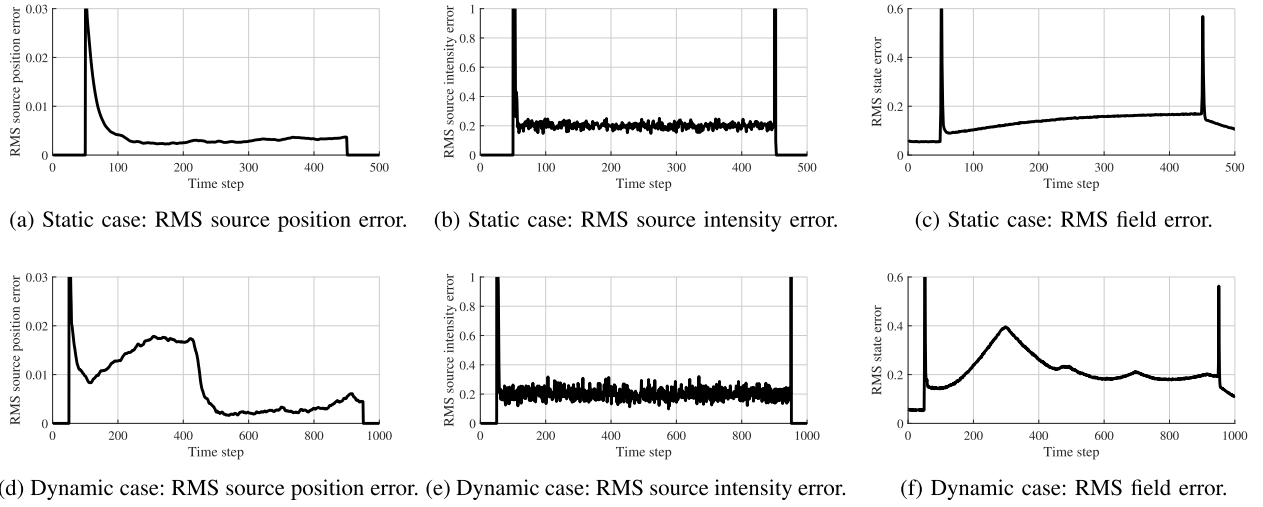


Fig. 6. Simulation results in the case of static source (a,b,c) and dynamic source (d,e,f).

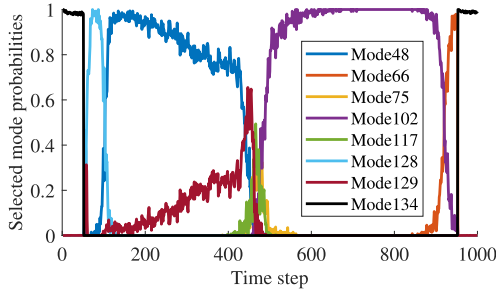


Fig. 7. Probabilities of selected modes (134 corresponds to the no-source mode) versus time.

Polynomial Matrix Description (PMD)

$$\begin{bmatrix} z\mathbf{I} - \mathbf{A}_{11} & \mathbf{B}_1(\mathbf{p}^0) & \mathbf{B}_1(\bar{\mathbf{p}}^0) \\ \mathbf{0} & \Delta(z) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Delta(z) \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1,k} \\ u_k \\ \bar{u}_k \end{bmatrix} = \mathbf{0}$$

$$\mathbf{y}_k = \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{1,k} \\ u_k \\ \bar{u}_k \end{bmatrix}$$

where $\Delta(z) = \det(z\mathbf{I} - \mathbf{F})$. In fact, we can now use the PBH observability test for PMDs (Antsaklis & Michel, 2006, p. 562) in order to conclude that (48) is observable if and only if

$$\text{rank} \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} & \mathbf{0} \\ z\mathbf{I} - \mathbf{A}_{11} & \mathbf{B}_1(\mathbf{p}^0) & \mathbf{B}_1(\bar{\mathbf{p}}^0) \\ \mathbf{0} & \Delta(z) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Delta(z) \end{bmatrix} = n_o + 2$$

for any $z \in \mathbb{C}$. In turn, the latter condition is equivalent to requiring that (22) holds for any $z \in \text{sp}\{\mathbf{F}\}$ (recall that the roots of $\Delta(z)$ are the eigenvalues of $\mathbf{F}$). $\square$

Proof of Theorem 2. Consider a pair of distinct elements $\mathcal{E}_j, \mathcal{E}_\ell$ and suppose that condition (24) holds for some z . Clearly, this implies that

$$\text{rank} [\mathbf{B}_1(\mathcal{E}_j), \mathbf{B}_1(\mathcal{E}_\ell)] = |\mathcal{Y}_j \cup \mathcal{Y}_\ell|. \quad (49)$$

Notice also that, for any $\mathbf{p}^0 \in \mathcal{E}_j$, one has $\mathbf{B}_1(\mathbf{p}^0)u_k = \mathbf{B}_1(\mathcal{E}_j)\eta_k$ with $\eta_k = \text{col}(\omega_i(\mathbf{p}^0)u_k, i \in \mathcal{Y}_j)$. Then, for any pair of source locations $\mathbf{p}^0, \bar{\mathbf{p}}^0$ and any pair of intensity sequences $u, \bar{u}$, Eq. (47) can be

rewritten as

$$\begin{aligned} \mathbf{x}_{1,k+1} &= \mathbf{A}_{11}\mathbf{x}_{1,k} + \mathbf{B}_1(\mathcal{E}_j)\eta_k - \mathbf{B}_1(\mathcal{E}_\ell)\bar{\eta}_k \\ \mathbf{y}_k &= \mathbf{C}_1\mathbf{x}_{1,k} \end{aligned} \quad (50)$$

with $\eta_k = \omega(\mathbf{p}^0)u_k$ and $\bar{\eta}_k = \omega(\bar{\mathbf{p}}^0)\bar{u}_k$.

In order to prove fact (i), notice that, when condition (24) holds for any $z \in \mathbb{C}$, then the output trajectory of the system (50) can be zero for any k if and only if

$$\mathbf{B}_1(\mathcal{E}_j)\eta_k = \mathbf{B}_1(\mathcal{E}_\ell)\bar{\eta}_k \quad (51)$$

for any k . We show now that, in view of (49), condition (51) holds if and only if the sources are identically null, i.e. $u_k = \bar{u}_k = 0$ for any k , or the sources are the same, i.e. $\mathbf{p}^0 = \bar{\mathbf{p}}^0$ and $u_k = \bar{u}_k$ for any k . To this end, notice that $\mathbf{B}_1(\mathcal{E}_j)$ is a $n_o \times (d+1)$ matrix whose columns are the vectors $\mathbf{B}_1(\mathbf{p}_i)$ for $i \in \mathcal{Y}_j$. Thus, condition (51) can be rewritten as

$$\sum_{i \in \mathcal{Y}_j} \mathbf{B}_1(\mathbf{p}_i) \omega_i(\mathbf{p}^0) u_k = \sum_{i \in \mathcal{Y}_\ell} \mathbf{B}_1(\mathbf{p}_i) \omega_i(\bar{\mathbf{p}}^0) \bar{u}_k$$

or, equivalently, as

$$\begin{aligned} \sum_{i \in \mathcal{Y}_j \setminus \mathcal{Y}_\ell} \mathbf{B}_1(\mathbf{p}_i) \omega_i(\mathbf{p}^0) u_k - \sum_{i \in \mathcal{Y}_\ell \setminus \mathcal{Y}_j} \mathbf{B}_1(\mathbf{p}_i) \omega_i(\bar{\mathbf{p}}^0) \bar{u}_k \\ + \sum_{i \in \mathcal{Y}_j \cap \mathcal{Y}_\ell} \mathbf{B}_1(\mathbf{p}_i) [\omega_i(\mathbf{p}^0) u_k - \omega_i(\bar{\mathbf{p}}^0) \bar{u}_k] = 0. \end{aligned}$$

The overall number of elements of the three summations correspond to the number of distinct vertices in $\mathcal{Y}_j \cup \mathcal{Y}_\ell$. For the reader's convenience, an example pertaining to a 2-dimensional domain is provided in Fig. 8. When condition (49) holds, then the vectors $\mathbf{B}_1(\mathbf{p}_i)$ with $i \in \mathcal{Y}_j \cup \mathcal{Y}_\ell$ are linearly independent. This implies that, when the elements are not adjacent $\mathcal{Y}_j \cap \mathcal{Y}_\ell = \emptyset$, the above condition can be satisfied if and only if $u_k = \bar{u}_k = 0$ because at least one $\omega_i(\mathbf{p}^0)$ and one $\omega_i(\bar{\mathbf{p}}^0)$ is different from 0. When instead the elements are adjacent, i.e., $\mathcal{Y}_j \cap \mathcal{Y}_\ell \neq \emptyset$, then the above condition can be satisfied if and only if

$$\omega_i(\mathbf{p}^0) u_k = 0 \quad \text{for } i \in \mathcal{Y}_j \setminus \mathcal{Y}_\ell \quad (52)$$

$$\omega_i(\bar{\mathbf{p}}^0) \bar{u}_k = 0 \quad \text{for } i \in \mathcal{Y}_\ell \setminus \mathcal{Y}_j \quad (53)$$

$$\omega_i(\mathbf{p}^0) u_k - \omega_i(\bar{\mathbf{p}}^0) \bar{u}_k = 0 \quad \text{for } i \in \mathcal{Y}_j \cap \mathcal{Y}_\ell. \quad (54)$$

In order to exclude the trivial solution $u_k = \bar{u}_k \neq 0$ for any k , suppose now that there exists at least one time instant k in which at least one between u_k and $\bar{u}_k$ is different from 0. Without loss of generality, we can take $u_k \neq 0$. Then, condition (52) can be

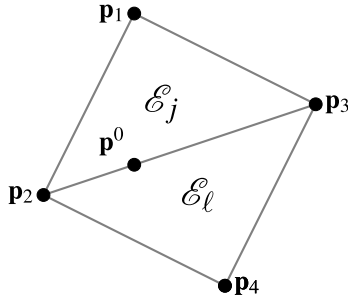


Fig. 8. Two elements in a 2-dimensional domain sharing a boundary. In this case, $\mathcal{V}_j = \{1, 2, 3\}$ and $\mathcal{V}_\ell = \{2, 3, 4\}$. Then, the union $\mathcal{V}_j \cup \mathcal{V}_\ell$ can be partitioned into the three sets $\mathcal{V}_j \setminus \mathcal{V}_\ell = \{1\}$, $\mathcal{V}_\ell \setminus \mathcal{V}_j = \{4\}$, and $\mathcal{V}_j \cap \mathcal{V}_\ell = \{2, 3\}$. A point $\mathbf{p}^0$ located on the shared boundary belongs to $\mathcal{E}_j$ with coefficients $\omega_1(\mathbf{p}^0) = 0$, $\omega_2(\mathbf{p}^0)$, and $\omega_3(\mathbf{p}^0)$ and belongs also to $\mathcal{E}_\ell$ with coefficients $\omega_2(\mathbf{p}^0)$, $\omega_3(\mathbf{p}^0)$, and $\omega_4(\mathbf{p}^0) = 0$. This means that a source located on the shared boundary between two elements $\mathcal{E}_j$ and $\mathcal{E}_\ell$ can be represented in two different ways as part of $\mathcal{E}_j$ or as part of $\mathcal{E}_\ell$. The two representations are equivalent and clearly lead to the same outputs. This is not a problem because the objective is not to identify in which element is the source, but rather to identify the source position $\mathbf{p}^0$ and intensity u_k .

satisfied only if $\omega_i(\mathbf{p}^0) = 0$ for all $i \in \mathcal{V}_j \setminus \mathcal{V}_\ell$. This means that the source $\mathbf{p}^0$ must be located on the shared boundary between the elements $\mathcal{E}_j$ and $\mathcal{E}_\ell$ (recall that $\omega_i(\mathbf{p}^0)$ with $i \in \mathcal{V}_j$ defines a convex combination). Then, at least one of the $\omega_i(\mathbf{p}^0)$ for $i \in \mathcal{V}_j \cap \mathcal{V}_\ell$ must be different from 0. As a consequence, condition (54) can be satisfied only if $\bar{u}_k \neq 0$. Then, from (53) we must have $\omega_i(\bar{\mathbf{p}}^0) = 0$ for $i \in \mathcal{V}_\ell \setminus \mathcal{V}_j$, i.e., also the source $\bar{\mathbf{p}}^0$ must be located on the shared boundary between the elements $\mathcal{E}_j$ and $\mathcal{E}_\ell$. Notice now that, by summing condition (54) over i , we obtain

$$\begin{aligned} 0 &= \sum_{i \in \mathcal{V}_j \cap \mathcal{V}_\ell} [\omega_i(\mathbf{p}^0) u_k - \omega_i(\bar{\mathbf{p}}^0) \bar{u}_k] \\ &= u_k \sum_{i \in \mathcal{V}_j \cap \mathcal{V}_\ell} \omega_i(\mathbf{p}^0) - \bar{u}_k \sum_{i \in \mathcal{V}_j \cap \mathcal{V}_\ell} \omega_i(\bar{\mathbf{p}}^0) = u_k - \bar{u}_k. \end{aligned}$$

Then, condition (54) can be satisfied only if the two intensities u_k and $\bar{u}_k$ are equal. Hence, by again exploiting (54), we get

$$\omega_i(\mathbf{p}^0) = \omega_i(\bar{\mathbf{p}}^0)$$

for $i \in \mathcal{V}_j \cap \mathcal{V}_\ell$. In turn, this implies,

$$\mathbf{p}^0 = \sum_{i \in \mathcal{V}_j \cap \mathcal{V}_\ell} \omega_i(\mathbf{p}^0) \mathbf{p}_i = \sum_{i \in \mathcal{V}_j \cap \mathcal{V}_\ell} \omega_i(\bar{\mathbf{p}}^0) \mathbf{p}_i = \bar{\mathbf{p}}^0.$$

Summing up, conditions (52)–(54) [or, equivalently, condition (51)] hold only when $u_k = \bar{u}_k = 0$ or $u_k = \bar{u}_k$ and $\mathbf{p}^0 = \bar{\mathbf{p}}^0$. This concludes the proof of fact (i).

As for fact (ii), consider the following system of linear equations

$$\begin{bmatrix} z\mathbf{I} - \mathbf{A}_{11} & \mathbf{B}_1(\mathcal{E}_j) & \mathbf{B}_1(\mathcal{E}_\ell) \\ \mathbf{C}_1 & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \eta \\ -\bar{\eta} \end{bmatrix} = \mathbf{0}$$

which, by exploiting the structure of $\mathbf{B}_1(\mathcal{E}_j)$ and $\mathbf{B}_1(\mathcal{E}_\ell)$, can be also written as

$$\begin{aligned} \mathbf{C}_1 \mathbf{x}_1 &= 0 \\ (z\mathbf{I} - \mathbf{A}_{11}) \mathbf{x}_1 + \sum_{i \in \mathcal{V}_j} \mathbf{B}_1(\mathbf{p}_i) \eta_i - \sum_{i \in \mathcal{V}_\ell} \mathbf{B}_1(\mathbf{p}_i) \bar{\eta}_i &= 0. \end{aligned}$$

By decomposing $\mathcal{V}_j \cup \mathcal{V}_\ell$ as before, the above system of linear equations can be rewritten as

$$\mathbf{C}_1 \mathbf{x}_1 = 0$$

$$\begin{aligned} (z\mathbf{I} - \mathbf{A}_{11}) \mathbf{x}_1 + \sum_{i \in \mathcal{V}_j \setminus \mathcal{V}_\ell} \mathbf{B}_1(\mathbf{p}_i) \eta_i - \sum_{i \in \mathcal{V}_\ell \setminus \mathcal{V}_j} \mathbf{B}_1(\mathbf{p}_i) \bar{\eta}_i \\ + \sum_{i \in \mathcal{V}_j \cap \mathcal{V}_\ell} \mathbf{B}_1(\mathbf{p}_i) [\eta_i - \bar{\eta}_i] &= 0. \end{aligned}$$

Supposing that the rank condition (24) holds for $z \in \text{sp}\{\mathbf{F}\}$, then for such values of z the only solutions of the above system of equations are of the form

$$\mathbf{x}_1 = 0 \quad (55)$$

$$\eta_i = 0 \quad \text{for } i \in \mathcal{V}_j \setminus \mathcal{V}_\ell \quad (56)$$

$$\bar{\eta}_i = 0 \quad \text{for } i \in \mathcal{V}_\ell \setminus \mathcal{V}_j \quad (57)$$

$$\eta_i - \bar{\eta}_i = 0 \quad \text{for } i \in \mathcal{V}_j \cap \mathcal{V}_\ell. \quad (58)$$

When $\eta_i = \omega(\mathbf{p}^0)u$ and $\eta_i = \omega(\bar{\mathbf{p}}^0)\bar{u}$, as previously shown the only solutions of the above system of equations are of the form $u = \bar{u} = 0$ or $\mathbf{p}^0 = \bar{\mathbf{p}}^0$ and $u = \bar{u}$. This means that, for all z for which the rank condition (24) holds, when $\mathbf{p}^0 \neq \bar{\mathbf{p}}^0$ the system of linear equations

$$\mathbf{C}_1 \mathbf{x}_1 = 0$$

$$(z\mathbf{I} - \mathbf{A}_{11}) \mathbf{x}_1 + \mathbf{B}_1(\mathbf{p}^0)u - \mathbf{B}_1(\bar{\mathbf{p}}^0)\bar{u} = 0$$

has the only solution $\mathbf{x}_1 = 0$, $u = 0$, and $\bar{u} = 0$. Hence, the rank condition of Theorem 1 fact (ii) holds and the source is identifiable. $\square$

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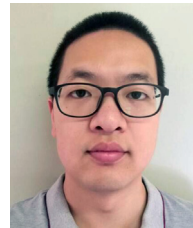
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